

PRATHAM SAURABH

(630) 881-5247 | psaurabh6@gatech.edu | [linkedin.com/in/pratham-saurabh](https://www.linkedin.com/in/pratham-saurabh) | prathamsaurabh.com

EDUCATION

Georgia Institute of Technology

BS Computer Science, Concentration in Artificial Intelligence and Cyber Security

GPA: 3.7

Graduation: May 2027

- **Relevant Coursework:** Computer Architecture, Data Structures, Algorithms, Embedded Systems, Systems Reliability Hardware/Software Integration, Linear Algebra, Statistical Inferences, Machine and Deep Learning, Communication Skills
- **Languages:** Python (NumPy, Pandas), Java, JavaScript/TypeScript, C++, Go, Rust, Kotlin, SQL, C, HTML, CSS, Angular
- **Software:** React, Node.js, Next.js, Docker, GitHub Actions, AWS, Linux, TensorFlow, PyTorch, Scikit-learn, Jira/Confluence, Figma
- **Clubs:** Georgia Tech IOS club – Bootcamp, Create X, Ivan Allen Student Advisory Board, Vertically Integrated Projects

PROFESSIONAL EXPERIENCE

Lockheed Martin

Full-Stacks Software Engineering Intern

Marietta, GA

June 2025 – August 2025

- Developed validation pipelines in Python, applying feature engineering to improve detection accuracy in flight data
- Co-Authored a Software Assurance Plan, defining documentation standards and coding guidelines to improve code reviews
- Deployed compliance workflows at scale across 12+ avionics systems, enhancing security in distributed environments
- Built C++ modules integrated with avionics systems, analyzing system logs to anomalies and improve compliance testing
- Engineered scalable Python systems to process avionics data at high volume, optimizing reliability and edge-case coverage

Systems Engineering Intern (AMMM Cybersecurity)

May 2024 – August 2024

- Developed secure Python pipelines with static analysis, enhancing compliance and software reliability in defense systems
- Utilized NIST 800-53 and CNSSI No. 1253 controls to refine security policies for 50+ aviation subsystems reducing audit gaps
- Refactored Ada based legacy models improving modularity and maintainability, reducing static analysis warnings by 30%
- Collaborated with Avionics teams to integrate findings into C-130J maintenance systems, decreasing redundant review loops
- Applied statistical models in Python to analyze avionics data, validating reliability and reducing redundant manual reviews

Georgia Institute of Technology Research

Undergraduate Machine Learning/Artificial Intelligence Researcher (STEMcomm)

Atlanta, GA

January 2025 – Present

- Conducted research on Artificial Intelligence, data collection methods, and physics applications to advance STEM education (R&D)
- Automated data scraping pipelines (Selenium), delivering clean structured datasets for reuse across multiple projects
- Integrated APIs and databases (Postgres) into production-ready research prototypes, enhancing scalability for collaborators
- Built end-to-end AI features with PyTorch and React/TypeScript, iterating on UIs and ML models for large datasets.

University of Georgia Research

Undergraduate Hardware/GPU Researcher

Athens, GA

June 2022 – May 2024

- Developed an API security interface framework for IoT systems enhancing security using APK files firmware (APKtools)
- Conducted GPU hardware validation and debugging, ensuring accuracy, performance reliability, and engineering robustness
- Analyzed and tested PLC devices, applying embedded systems knowledge to improve reliability in production-ready code
- Implemented reusable C/C++ components and debugged embedded algorithms, ensuring efficiency under large workloads

ZenFrenz

Data Engineering/Marketing Intern

Remote

June 2023 – August 2023

- Presented data-driven insights to the CEO, applying problem solving strategies that boosted profit by 12% (eCommerce)
- Identified merchandising purchases from 5000+ customers data across Amazon FBA and Shopify, improving targeted precision
- Built decision-tree models for customer engagement, mapping features and validating predictions to guide marketing strategy

PROJECTS

PhysicsGPT

STEMcomm

- Developed transformer-based Physics model in Python, performing feature engineering and evaluating model performance
- Pushed production-level transformer workflows with Python/NLP, addressing edge cases and improving accuracy for users
- Leveraged NLP techniques to enhance response accuracy, ensuring reliable and detailed answers for complex physics queries

GeNN CI/CD Workflow Integration

INCF/GSOC 2025

- Built CI/CD pipelines for GPU simulations, validating model performance metrics and automating statistical testing
- Designed experiments to validate ML workflows, analyzing performance metrics and ensuring reliable deployment pipelines
- Validated workflows on Ubuntu runners, debugging cross-platform jobs to ensure scalable, secure, and reliable deployments

ADDITIONAL

Interests: Cooking, Hiking, Exploring, Basketball, Football, Tennis (Highest UTR 7), Gardening, DIY Builds, Hapkido

Security Clearance: Interim Secret (Debriefed)

Languages: English (Fluent), Hindi (Proficient in Speaking), Spanish (Intermediate in Speaking)